

Hassan Elhllos

Embedded Systems & PCB Design Engineer

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PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in Embedded Systems, PCB Design, IoT, and Industrial Automation. 3+ years of freelance experience developing custom hardware and smart devices end-to-end, from schematic and PCB design to C/C++ firmware, connectivity, and prototype validation. Hands-on experience with product development, reverse engineering, BOM sourcing, and manufacturing coordination. 1st Place winner at Suez Canal University's Canal Startup Sprint with a custom industrial PLC.

TECHNICAL SKILLS

Embedded Systems: ESP32, ESP12E/F, STM32, Arduino, C/C++, GPIO, Timers, Interrupts, Firmware Development

PCB and Hardware Design: KiCad, Altium Designer, Schematic Design, PCB Layout, PCB Routing, EMC-aware Design, Reverse Engineering, BOM Management, Soldering (THT/SMD)

Industrial Automation: Siemens S7-1200, Schneider PLC, Ladder Logic, Modbus RTU (RS485), VFD Configuration, Classic Control, Star-Delta, Motor Protection

Connectivity and Protocols: MQTT, HTTP, UART, SPI, I2C, RS485, Wi-Fi, BLE, ESP RainMaker, Blynk, HiveMQ

Software and Tools: Python, SQL, Git/GitHub, Ladder Diagram, Proteus, Dashboard Development, Oscilloscope, Multimeter, Hardware Troubleshooting

Computer Vision: Raspberry Pi, OpenCV, Object Detection, Underwater Vision

EXPERIENCE

Independent IoT & Hardware Developer Freelance

2023 – Present

- Develop custom IoT products and complex PCB designs for client and academic projects, covering hardware, firmware, connectivity, and prototype validation.
- Handle end-to-end product development tasks including reverse engineering, BOM preparation, supplier coordination, and manufacturing readiness.

Industrial Automation Trainee HA Consulting Group

Sep 2025 – Nov 2025

- Configured VFDs and designed classic control circuits including star-delta starters and overload protection.
- Programmed Siemens S7-1200 and Schneider PLC systems using industrial control logic.

Advanced Industrial Electronics & PCB Diploma Digital Fabrication — High-Speed PCB Design, Signal Integrity & SMPS

Dec 2025 – Feb 2026

- Advanced PCB design covering high-speed layout, impedance control, signal integrity, differential pairs, multi-layer Altium workflows, and SMPS design.

IoT Intern Creativa Innovation Hub (ITIDA) — Grade: Excellent

May 2025 – Jun 2025

- Configured ESP32 edge platforms, established MQTT communication, and developed cloud dashboards for real-time monitoring.
- Architected a Smart Agriculture system connecting field devices to cloud services through edge-to-cloud MQTT communication.

Computer Vision Volunteer SCU Racing Team (ROV), Suez Canal University

2024 – 2025

- Developed underwater object-detection and navigation modules using Python and OpenCV on Raspberry Pi.

Data Analysis Intern Information Technology Institute (ITI)

Jul 2024 – Aug 2024

- Applied SQL, data cleaning, visualization, and dashboards to extract and communicate insights from datasets.

SELECTED PROJECTS

Industrial PLC Controller: Designed a locally manufactured ESP32-based PLC with EMC-aware PCB, 8 inputs/8 outputs, Modbus communication for HMI/SCADA, custom Ladder-programming software, and 3D-printed enclosure; **1st Place** at Canal Startup Sprint.

Flomac Automatic Pump Controller: Reverse-engineered a proprietary control board and reproduced its schematic, BOM, and PCB in KiCad; validated a functional prototype and coordinated manufacturing/sourcing for production targeting 1,000 units, with a planned scale to 20,000 units.

Custom Smart Home System: Developed an ESP32-based controller with an 8-channel relay interface on a custom KiCad PCB, supporting remote access and automation through ESP RainMaker/Blynk.

Smart Health & Activity Watch: Built an ESP32 wearable prototype with touchscreen UI and biometric sensing for heart rate, SpO2, and temperature using I2C/SPI peripherals.

Industrial IoT Systems: Built ESP32-based monitoring and control systems for cold-storage panels and conventional motor-control panels, adding remote monitoring, control, and fault alerts.

EDUCATION

B.Sc. in Electrical Engineering Suez Canal University, Faculty of Engineering

2021 – 2026

Diploma in Information Technology Advanced Technical School – Ismailia, Egypt

2016 – 2021

HONORS AND CERTIFICATIONS

1st Place — Canal Startup Sprint, Suez Canal University, with the Industrial Controller (ESP32 PLC) graduation project.

Certifications: Industrial Automation (HA Consulting Group); IoT Internship – Excellent (Creativa Innovation Hub / ITIDA); Data Analysis (ITI).

LANGUAGES

Arabic (Native) and English (Professional)